

Problem Statement:

You are working as a data analyst for a marketing team that is studying consumer preferences related to product color and social media behavior.

1. The company claims that the distribution of colors in their newly launched candy pack follows the proportions:
 - Red: 30%, Blue: 25%, Green: 20%, Yellow: 15%, and Orange: 10%.
2. You purchase a sample of 100 candies from the market and count the actual number of candies of each color:
 - Red: 28, Blue: 30, Green: 20, Yellow: 12, Orange: 10.
3. Using this data, perform a **Chi-Square Goodness of Fit Test** to determine whether the observed color distribution significantly differs from the company's claim.
Use a 5% level of significance. Clearly state the null and alternative hypotheses, perform the test, and interpret the result.

In the same study, a survey was conducted among 100 students to examine whether **age group** influences the **preferred social media platform**. The responses are summarized in the contingency table below:

Platform / Age Group	15–20	21–25	26–30
Instagram	20	18	7
Twitter	10	15	8
Facebook	5	8	9
LinkedIn	2	4	4

Perform a **Chi-Square Test of Independence** to determine whether there is a significant association between age group and preferred social media platform.

Instructions:

- Use Python and relevant libraries such as scipy.stats, pandas, and numpy to perform both tests.
- Add visualizations (e.g., bar plots, heatmaps) wherever appropriate.